

Keepsake — Build Work Log

Per-phase summary of work, decisions, and changes

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Overview

This is a breadth-first **2.5D working prototype** of Keepsake, built across the Design Bible v3 phases (A–F). The goal is a coherent, testable experience that touches as many systems as possible rather than finishing any one to production depth. Everything runs in the browser (React 19 + TypeScript + Vite, Tailwind v4), persisted to **IndexedDB**; there is no backend yet, so account, storage, and multi-user systems are simulated *locally* and labelled as such.

Art direction. The room and chrome draw on *Little Big Planet* (handcrafted, tactile "pop-out" objects), *Sly Cooper* (storybook, bold stylised shapes, warm noir), and *What Remains of Edith Finch* (a cozy, slightly melancholic diorama house, warm light, drifting dust). It is realised in 2.5D with layered parallax, painterly gradients, grain, and soft vignette — the "2.5D over full 3D" open decision from Bible §4/§8.

Ideas that *deviate* from the Bible are collected in the companion "Suggestions & Deviations" PDF; quick, obvious fixes and any clear privacy/security issues are acted on directly and noted here.

Architecture at a glance

- `store/appStore` — one app state (profile, books, archive, environment, social, pins, timeline, achievements) with debounced IndexedDB autosave.
- `store/nav` — a small room-view router plus the local *View as* mode.
- `hooks/useScrapbook` — editor state over the active book.
- The room is the interface: a diorama hub whose objects open each system (Bible §2/§8).

PHASE A

Core scrapbook

Built

- Book → open two-page spread on a wooden desk; cream paper, spine shadow, drop shadows.
- Photographs as tactile prints (polaroid / tape / flush frames); select, drag, rotate, resize, restack, replace, remove.
- Positions stored as page percentages so layout holds across screen sizes (Bible requirement).

Notes

- Data model follows the Bible's `Scrapbook → Page → PageElement` shapes.

PHASE B Scrapbook MVP

Built

- Multi-photo pages (up to 6), z-order, sensible initial positions.
- Editable handwritten captions (character-limited) with sharpie ink colours.
- **Layout presets** — Grid / Column / Scatter for fast placement (§5).
- **Directional 3D page-turn** animation across the spine: the turning leaf shows the current page on the front and the *real incoming page* on the back, with a short Web Audio paper-rustle. Reduced-motion still does an instant swap and stays silent (§7).
- **Undo / redo** in the editor (40-step stack, burst-coalesced so a drag is one step) with chips and ⌘Z / Ctrl+Z, ⌘⇧Z / Ctrl+Y. ⓘ opens a shortcut list.
- **Stickers** as page elements sharing the photo/caption transform model (§11).
- **Free transform for photos & stickers:** a drag-to-rotate handle (hold Shift to snap to 15°) for any angle, plus **two-finger pinch to resize and twist to rotate** on touch devices (Pointer Events, so mouse/pen/multi-touch all work). Toolbar buttons remain for fine steps.

Changes / fixes

- Fixed two React StrictMode-exposed impurities discovered by testing (a preset counter and a page-turn commit that ran inside a state updater).

PHASE C Persistent product local

Built

- **Multiple books + bookshelf:** create, rename, choose leather cover, set visibility, delete; select a book by its cover (§9).
- **Filing cabinet / archive:** every uploaded photo is kept, searchable by category, favouritable; "place in book" adds a *reference* to the existing photo rather than duplicating it (§10).

- Per-book **visibility** (private / friends / public).

Simulated locally. Accounts, object storage, and server-side permission enforcement do not exist yet. Visibility is a local *preview* only (see Phase E and the "real vs simulated" section). Uploaded images are downscaled and stored as data URLs in IndexedDB.

PHASE D Room alpha (2.5D)

Built

- A diorama **room hub** with three parallax depth layers that respond to the pointer; interactive objects lift and reveal a handwritten label on hover.
- Authored "camera views" (room → book / shelf / archive / timeline / map / guest book) via a simple view router — a CameraDirector stand-in (§8).
- **Window / day-night / seasons / weather / lamp:** sky and light shift with time (auto by clock, or forced), a seasonal tint, rain/snow, and a warm lamp glow after dark (§14).
- Ambient **dust motes** and grain for the Edith-Finch mood.

Fixes

- Parallax layer wrappers were intercepting clicks; made them non-interactive so objects beneath remain clickable.

PHASE E Social local sim

Built

- **Visitor "View as"** (Close friend / Friend / Public) previews the room and library through a visitor's eyes; private books and the archive are hidden accordingly, and the book editor becomes read-only (§17/§18).
- **Guest book** — flat, non-threaded notes, no counters or reply chains (§19).
- **Page notes + approval + whispers** — a visitor can leave a short note on a spread; it stays hidden until the owner approves it; approved notes then occasionally drift into the room as a quiet, dismissible "whisper" (§19).

Simulated locally. There are no real other users; "View as" is a client-side preview and notes/visibility are enforced only in the UI. A real permission-inheritance engine and server enforcement are required before any actual sharing (§17).

Built

- **Timeline wall** — add year milestones on a vertical line (§15).
- **Memory atlas / map** — an illustrated map; click to pin a place + note; exact coordinates are never used (approximate/manual by default, §16).
- **Craft drawer / stickers** in the editor (§11).
- **Hidden achievements — event ledger** (§12): a pure rules engine derives completion from state, so grants are idempotent and reproducible; each is stamped with a completion time and a "seen" flag separates completion from reward presentation. Presented as gentle "keepsakes found"; a panel lists found vs undiscovered.
- **CRT music** — a provider-adaptor player (Bible §13): an offline, licence-free generative ambient pad (Web Audio), *and* a "**Connect Spotify**" account login via browser-safe OAuth PKCE (no client secret). Once connected it lists your playlists and can associate one per book; an embedded Spotify player handles playback (a manual playlist link works without login too). Activation needs a Spotify Developer app Client ID (`VITE_SPOTIFY_CLIENT_ID`) — requested via the environment setup panel; full in-app playback control needs Spotify Premium (Web Playback SDK).
- **Curated-RNG opportunity queue** (§20): sixteen candidates (welcome, reward, empty-page, whisper, photo-out, ajar-book, bookmark, guest-wave, map-glint, night-moth, pressed-flower, season-note, letter, window-draft, kettle, tea-ring) each carry eligibility, priority, and a cooldown. A curator picks at most one per session (highest priority; ties broken randomly; quieter events may present nothing) and records a persisted receipt. Force a kind with `?event=` . Everything is ambient and non-interrupting.
- **Keyboard / screen-reader pass** (§23, first cut): skip links, a `<main>` landmark, named icon buttons, focus rings on room objects, Tab-trapped dialogs that restore focus, a live spread counter, and tour copy exposed in full to assistive tech.
- **Onboarding** — first visit opens a *Messenger*-style dialogue tour (name plate + paper bubble + overlapping next) narrated by “the house.” It walks the room object by object (book, window, shelf, archive, timeline, map, guest book, CRT, and the top bar), with a spotlight cut-out, typewriter line, skip (“wander off” / Escape), and replay from room settings or `?tour=1` . Completing or skipping sets `progress.completedTour` so it does not auto-start again. This is a requested deviation from Bible §21 “discovery over instruction” — kept skippable and house-voiced so it still feels like the room speaking, not a checklist.
- **Export / print** — a print-ready view of a book; "Save as PDF" in the browser produces a shareable keepsake (§25, software half).
- **Offline read mode** — a service worker + web manifest cache the app shell and assets (production build), and scrapbook data already lives in IndexedDB, so a loaded room can be revisited offline and installed to the home screen (§22).

Deviations acted on during the build

Change	Why	Bible
Persist photos (as data URLs) so books survive a refresh	Much better to actually test with; small change	§3 (prototype images were "intentionally temporary")
Client-side visibility gating for private books / archive and "View as"	A clear privacy expectation; implemented as an honest local preview and flagged that real enforcement is server-side	§17
Downscale uploads to ≤1400px	Protects browser memory / storage quota	§6/§22 (listed risks)

Anything larger or subjective was *not* acted on — it lives in the Suggestions & Deviations PDF for your review.

Real vs simulated (please read)

- **Real & local:** the whole editor, room, environment, library, archive, timeline, map, guest book, achievements, music, export — all working in the browser and saved to IndexedDB.
- **Simulated (no backend yet):** accounts, cloud photo storage, multi-user visiting, and *server-side privacy enforcement*. "View as" and visibility are previews, not security. This is the top risk area in the Bible (§17) and needs a real backend before any sharing is enabled.

Testing

- `npm run build` (tsc + vite) passes with no type errors.
- Browser-verified end to end: room navigation to every object; open/edit book (photos, captions, stickers, presets, page turns, export); day → night in the window; bookshelf, archive (search/favourite/place-in-book), timeline, map (pin), guest book; "View as" gating.
- No red console errors. Icon-only controls now carry accessible names; dialogs trap focus.

Where this could go next

- A real backend (Postgres + auth + object storage) and a proper permission-inheritance engine, turning the local "View as"/visibility previews into real, enforced privacy (§17).
- Commission illustrated parallax plates for the room — painted 2.5D first, not a Three.js rewrite. See the 3D art brief in the suggestions PDF.
- A real-device VoiceOver / TalkBack pass, and licensed sticker/craft packs.
- Spotify Web Playback SDK once a Client ID is injected on new agents (Premium).